

Machine Learning Assignment 1 Report

Polynomial Regression

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1. Objective

The objective of this assignment is to develop polynomial regression models for two personalised datasets and predict the continuous target variable y for the corresponding test datasets. The first problem, var1, contains six input variables (x_1 to x_6), while the second problem, var2, contains three input variables (x_1 to x_3). The main task is to select an appropriate polynomial degree so that the model captures the underlying relationship without unnecessary overfitting.

2. Dataset Description

Problem	Training	Test	Input Variables	Target
var1	1000 rows	1000 rows	x_1 to x_6	y
var2	1000 rows	1000 rows	x_1 to x_3	y

3. Methodology

The model uses a three-step pipeline. First, PolynomialFeatures generates polynomial and interaction terms up to the selected degree. A degree- d polynomial includes terms whose total power is at most d . Second, StandardScaler standardises the generated features. Finally, Ridge regression is used to estimate the model coefficients. Ridge regularisation helps control the effect of the large number of correlated features produced by higher-degree polynomial expansion.

The polynomial degree was selected using 5-fold cross-validation on the training data only. For each candidate degree, the training data was split into five folds. Four folds were used for training and the remaining fold for validation, with the process repeated for all five folds. The mean validation MSE was then calculated. The degree with the lowest mean MSE was selected. The test datasets were not used during degree selection.

The Ridge regularisation strength was fixed at $\alpha = 30$ for var1 and $\alpha = 2$ for var2 during the degree-selection sweep.

4. Results for var1

Polynomial degrees from 1 to 10 were evaluated for var1. The validation MSE decreased as the degree increased from 1 to 5, reaching its minimum at degree 5. After degree 5, the validation MSE increased, so higher degrees did not improve the model's validation performance.

Degree	CV MSE	CV R^2
1	9.7977	0.0880
2	3.1033	0.7112
3	1.0287	0.9043
4	0.7437	0.9309
5	0.5275	0.9508
6	0.6036	0.9438
7	0.7211	0.9327
8	0.8710	0.9187

9	1.0866	0.8987
10	1.2832	0.8802

Selected model for var1: Polynomial degree 5 with Ridge $\alpha = 30$. The 5-fold cross-validation MSE is 0.527455 and the mean cross-validation R^2 is 0.950820.

5. Results for var2

Polynomial degrees from 1 to 20 were evaluated for var2. The validation MSE decreases substantially as the degree increases, reaching a minimum at degree 12. After degree 12, the validation MSE begins to increase. Although degree 8 already gives a much lower error than the lower-degree models, degree 12 has the lowest cross-validated MSE and was therefore selected.

Degree	CV MSE	CV R^2
1	35.4698	0.2219
2	20.8344	0.5376
3	11.2156	0.7476
4	3.5138	0.9229
5	1.6558	0.9637
6	0.7172	0.9841
7	0.4863	0.9893
8	0.3229	0.9928
9	0.2882	0.9936
10	0.2567	0.9942
11	0.2566	0.9943
12	0.2520	0.9944
13	0.2583	0.9942
14	0.2609	0.9942
15	0.2688	0.9940
16	0.2741	0.9939
17	0.2819	0.9937
18	0.2891	0.9936
19	0.2967	0.9934
20	0.3056	0.9932

Selected model for var2: Polynomial degree 12 with Ridge $\alpha = 2$. The 5-fold cross-validation MSE is 0.251993 and the mean cross-validation R^2 is 0.994355.

6. Final Models and Prediction

Problem	Features	Degree	Ridge α	CV MSE	CV R^2
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var1	x1 to x6	5	30	0.5275	0.9508
var2	x1 to x3	12	2	0.2520	0.9944

After selecting the final hyperparameters, each model was refitted using all 1,000 training rows for its respective problem. The trained models were then used to predict y for all 1,000 rows in the corresponding test dataset. The predictions were saved with a single column named y.

7. Conclusion

The cross-validation experiments resulted in a degree-5 polynomial Ridge model for var1 and a degree-12 polynomial Ridge model for var2. For var1, degree 5 gives the lowest validation MSE before the error increases at higher degrees. For var2, the error decreases up to degree 12 and then increases gradually. Therefore, the selected degrees provide the best validation performance among the tested polynomial degrees. Feature standardisation and Ridge regularisation were used to make the polynomial models more stable, particularly at higher degrees.

8. Deliverables

The completed prediction files are:

- BT2024113_pred_var1.csv
- BT2024113_pred_var2.csv

GitHub link - [Hasini-Samudrala/Polynomial-Regression](https://github.com/Hasini-Samudrala/Polynomial-Regression)